In [1]:

```
import numpy as np
import scipy as sp
import matplotlib.pyplot as plt
from tqdm import tqdm
```

Анализ детерминированной задачи

Определим вспомогательные функции.

In [2]:

```
def find_jacobian(
    func: callable,
    t: float,
    state: np.ndarray,
    params: dict,
    eps: float = 1e-6,
) -> np.ndarray:
    n = len(state)
    jacobian = np.zeros((n, n))

    for i in range(n):
        perturb = np.zeros(n)
        perturb[i] = eps

        f1 = func(t, state + perturb, **params)
        f2 = func(t, state - perturb, **params)
        jacobian[:, i] = (f1 - f2) / (2 * eps)

    return jacobian

def find_equilibrium(
    func: callable,
    guess: np.ndarray,
    params: dict,
) -> np.ndarray:
    def root_fn(state):
        return func(0, state, **params)

    solution, info, ier, _ = sp.optimize.fsolve(root_fn, guess, full_output=True)

    if ier == 1:
        return solution
    return None


def classify_equilibrium(
    eigenvalues: np.ndarray,
    eps: float = 1e-6
) -> str:
    real = np.real(eigenvalues)
    imag = np.imag(eigenvalues)

    if np.any(real > eps) and np.any(real < -eps):
        return 'saddle'
```

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```
    if np.all(np.abs(real) < eps):
        if np.any(np.abs(imag) > eps):
            return 'center'
        else:
            return 'degenerate'

    if np.any(np.abs(imag) > eps):
        if np.all(real < 0):
            return 'stable focus'
        elif np.all(real > 0):
            return 'unstable focus'

    if np.all(real < 0):
        return 'stable node'
    elif np.all(real > 0):
        return 'unstable node'

    return 'unknown'
```

Определим класс двумерной динамической системы с возможностью нахождения решения методом Рунге-Кутты 4 порядка с суммарной ошибкой на интервале решения, с возможностью построения бифуркационной диаграммы и фазового портрета системы, нахождения равновесий и предельных циклов.

In [ ]:

```
class DynamicSystem2D:

    def __init__(
        self,
        model_func: callable,
        params: dict = None,
    ) -> None:
        """
        Args:
            model_func: function returning a vector of (dx/dt, dy/dt)
            params: dictionary of parameters
        """
        self.model_func = model_func
        self.params = params if params is not None else {}

    def solve(
        self,
        state_init: np.ndarray,
        time_span: np.ndarray
    ) -> np.ndarray:
        """
        Find numeric solution for the system using 4-th order Runge-Kutta method.

        Args:
            state_init: initial state (x_0, y_0) of the dynamic system
            time_span: uniform time grid

        Returns:
            solution: array system states per time point throughout the time span
        """
        n = len(time_span)
```

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```
        h = time_span[1] - time_span[0]

        solution = np.zeros((n, len(state_init)))
        solution[0] = state_init.copy()

        for i in range(n - 1):
            t = time_span[i]
            state_curr = solution[i]

            k1 = h * self.model_func(t, state_curr, **self.params)
            k2 = h * self.model_func(t + h/2, state_curr + k1/2, **self.params)
            k3 = h * self.model_func(t + h/2, state_curr + k2/2, **self.params)
            k4 = h * self.model_func(t + h, state_curr + k3, **self.params)

            solution[i + 1] = state_curr + (k1 + 2 * k2 + 2 * k3 + k4) / 6

        return solution

    def plot_bifurcation_diagram(
        self,
        param_name: str,
        param_values: np.ndarray,
        state_init: np.ndarray,
        ax: plt.Axes,
    ) -> tuple[np.ndarray, np.ndarray]:
        """
        Plot bifurcation diagram for a dynamic system.

        Iterate over the dense grid of parameter values, and detect eigenvalue sign chan
        for Jacobians of equilibrium states, marking them as bifurcation points.

        Real part of the eigenvalues of the Jacobian indicates the system stability/inst
        per parameter value (<0 ==> stable, >0 ==> unstable).

        Returns:
            bifurcation_points: array of bifurcation.
            equilibria: array of equilibrium point coordinates.
        """
        equilibria = []
        stability_mask = []
        types = []
        param_used = []
        bifurcation_points = []

        eigenvalues_last = None

        for val in param_values:
            self.params[param_name] = val

            equilibrium = find_equilibrium(self.model_func, state_init, self.params)
            if equilibrium is None:
                continue

            jacobian = find_jacobian(self.model_func, 0, equilibrium, self.params)
            eigenvalues = np.linalg.eigvals(jacobian)

            # equillibrium classification
            eq_type = classify_equilibrium(eigenvalues)
            types.append(eq_type)
```

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```
        is_stable = np.all(np.real(eigenvalues) < 0)
        stability_mask.append(is_stable)

        # bifurcation check
        if eigenvalues_last is not None:
            if np.any(np.real(eigenvalues_last) * np.real(eigenvalues) < 0):
                bifurcation_points.append(val)
                ax.axvline(val, linestyle='--', label='bifurcation point')

        equilibria.append(equilibrium)
        param_used.append(val)

        eigenvalues_last = eigenvalues
        state_init = equilibrium

    equilibria = np.array(equilibria)
    stability_mask = np.array(stability_mask)
    param_used = np.array(param_used)
    bifurcation_points = np.array(bifurcation_points)

    x_eq = equilibria[:, 0]

    type_to_color = {
        'stable node': 'blue',
        'unstable node': 'red',
        'saddle': 'black',
        'stable focus': 'green',
        'unstable focus': 'orange',
        'center': 'purple',
        'degenerate': 'gray',
    }

    for t in set(types):
        mask = np.array([tt == t for tt in types])
        ax.scatter( param_used[mask], x_eq[mask], label=t, color=type_to_color.get(t

    ax.set_xlabel(param_name)
    ax.set_ylabel('x')
    ax.set_title('Bifurcation diagram (with equilibrium types)')
    ax.legend()
    ax.grid(True)

    return bifurcation_points, equilibria

def plot_trajectory(
    self,
    state_init: np.ndarray,
    time_span: np.ndarray,
    ax: plt.Axes,
    label: str,
) -> None:
    """
    Plot trajectory for given initial state.
    """
    solution = self.solve(state_init, time_span)
    ax.plot(solution[:, 0], solution[:, 1], label=label)
    ax.set_title('Phase portrait')
    ax.set_xlabel('x')
```

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```
        ax.set_ylabel('y')
        ax.legend(loc='upper right')
        ax.grid(True)

    def extract_limit_cycle(
        self,
        solution: np.ndarray,
        clip_ratio: float = 0.5,
        eps: float = 1e-4,
    ) -> np.ndarray | None:
        """
        Extract limit cycle from a trajectory if it exists.

        Returns:
            solution_ss: limit cycle (if found) or None
        """
        x = solution[:, 0]

        # clip transient (consider only the asymptotic behavior of the system)
        clip = int(len(x) * clip_ratio)
        solution_ss = solution[clip:]
        x_ss = solution_ss[:, 0]

        # check: is solution bounded
        if not np.all(np.isfinite(solution_ss)):
            return None

        # check: not an equilibrium (consider only cycles, not points)
        if np.var(x_ss) < eps:
            return None

        return solution_ss

    def plot_limit_cycle(
        self,
        ax: plt.Axes,
        time_span: np.ndarray,
        bounds: tuple = (-2, 2),
        n_attempts: int = 5,
    ) -> bool:
        """
        Try to detect and plot limit cycle.
        """
        found = False

        for _ in range(n_attempts):
            state_init = np.random.uniform(bounds[0], bounds[1], size=2)
            solution = self.solve(state_init, time_span)

            if not np.all(np.isfinite(solution)):
                continue

            cycle = self.extract_limit_cycle(solution)
            if cycle is not None:
                ax.plot(cycle[:, 0], cycle[:, 1], color='red', linewidth=2, label='limit
                found = True
                break

        return found
```

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```python
def plot_phase_portrait_dense(
    self,
    ax: plt.Axes,
    time_span: np.ndarray,
    bounds: tuple = (-2, 2),
    grid_size: int = 10,
    show_limit_cycle: bool = True,
) -> None:
    """
    Plot phase portrait based on a grid of initial states.
    """
    xmin, xmax = bounds
    ymin, ymax = bounds

    # initialize grid
    x_vals = np.linspace(xmin, xmax, grid_size)
    y_vals = np.linspace(ymin, ymax, grid_size)

    # test trajectories for all initial states on the grid
    for x0 in x_vals:
        for y0 in y_vals:
            state_init = np.array([x0, y0])

            solution = self.solve(state_init, time_span)
            if np.all(np.isfinite(solution)):
                ax.plot(solution[:, 0], solution[:, 1], color='blue', alpha=0.5)

    # plot vector field
    X, Y = np.meshgrid(np.linspace(xmin, xmax, 25), np.linspace(ymin, ymax, 25))
    U = np.zeros_like(X)
    V = np.zeros_like(Y)

    for i in range(X.shape[0]):
        for j in range(X.shape[1]):
            dx, dy = self.model_func( 0, np.array([X[i, j], Y[i, j]]), **self.params
            U[i, j] = dx
            V[i, j] = dy

    # normalize vector field
    norm = np.sqrt(U**2 + V**2)
    U /= (norm + 1e-8)
    V /= (norm + 1e-8)

    ax.streamplot(X, Y, U, V, density=1.2, color='black', linewidth=0.5)

    # try to find and plot the limit cycle
    if show_limit_cycle:
        found = self.plot_limit_cycle(ax=ax, time_span=time_span, bounds=bounds)
        if found:
            print("Limit cycle found!")

    ax.set_xlim(xmin, xmax)
    ax.set_ylim(ymin, ymax)
    ax.set_xlabel('x')
    ax.set_ylabel('y')
    ax.grid(True)
```

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1. Нейронная модель ФитцХью-Нагумо (FHN)

1. Найдем равновесия системы.

2. Исследуем равновесия на устойчивость.

Далее с помощью численной процедуры исследуем поведение собственных чисел якобиана в равновесных точках при различных значениях параметра .

In [4]:

```
def model_FHN(
    time: float,
    state: np.ndarray,
    delta: float = 0.1,
    a: float = 1,
):
    """
    FitzHugh-Nagumo neural model.
    Accepts `time` argument for interface consistency, but does not use it.

    Parameter restrictions:
    - delta << 1 (delta = 0.1)
    - a > 0
    """
    x, y = state
    return np.array([
        (x - (x ** 3) / 3 - y) / delta,
        x + a
    ])
```

In [5]:

```
FHN = DynamicSystem2D(
    model_FHN,
    params=dict(delta=0.1, a=1)
)

state_init = np.array([0, 0])
time_span = np.linspace(0, 50, num=2000)
param_values = np.linspace(0, 2, num=50)
```

3. Построим бифуркационную диаграмму по параметру .

In [6]:

```
fig, ax = plt.subplots()

bifurcation_points, equilibria = FHN.plot_bifurcation_diagram(
    param_name='a',
    param_values=param_values,
    state_init=state_init,
    ax=ax
)
```

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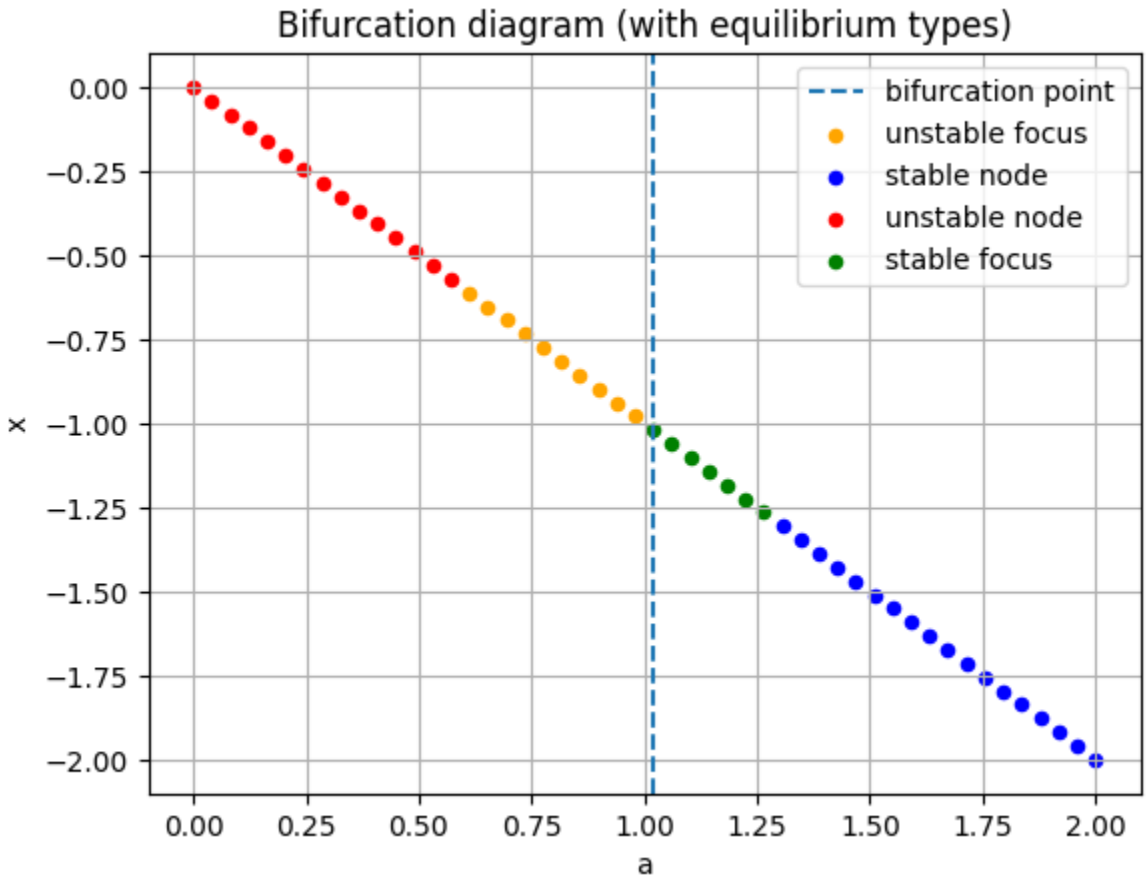
```
a_bif = bifurcation_points[0]
```



4. Построим типичные фазовые портреты для различных значений по бифуркационной диаграмме.

In [7]:

```
a_min = param_values[0]
a_max = param_values[-1]
a_values_to_types = [
    (0.25, "unstable node"),
    (0.75, "unstable focus"),
    (1.2, "stable focus"),
    (1.5, "stable node")
]
a_values, equilibrium_types = zip(*a_values_to_types)

equilibria_per_value = np.concatenate((param_values[:, None], equilibria), axis=1)
tolerance = abs(param_values[1] - param_values[0]) / 2

fig, axes = plt.subplots(1, len(a_values_to_types), figsize=(18, 4))

for ax, a, eq_type in zip(axes, a_values, equilibrium_types):
    FHN.params['a'] = a
    FHN.plot_phase_portrait_dense(
        ax=ax,
        time_span=time_span,
        bounds=(-2, 2),
        grid_size=10
    )
```

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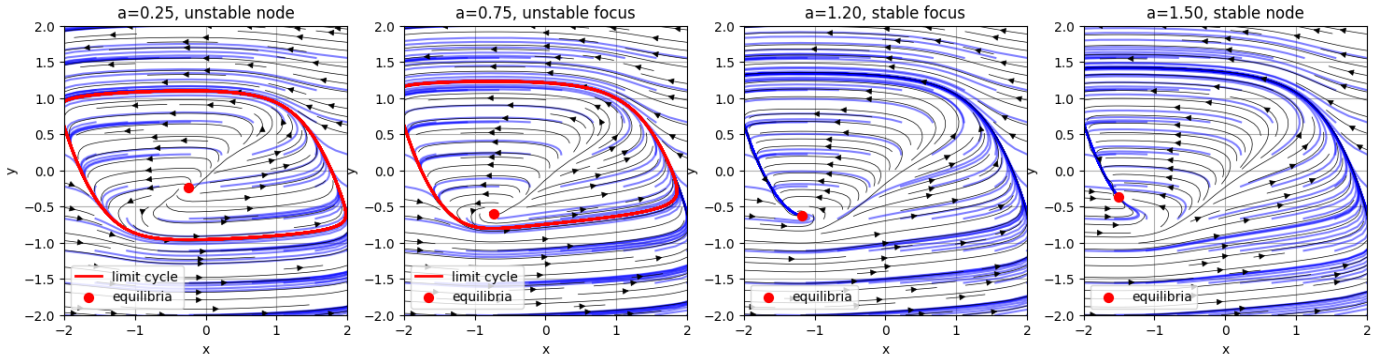
```
    equilibrium_idx = np.isclose(equilibria_per_value[:, 0], a, atol=tolerance)
    equilibrium_value = equilibria_per_value[equilibrium_idx][0]
    equilibrium_point = equilibrium_value[1:]

    ax.scatter(*equilibrium_point, color='red', marker='o', s=50, zorder=100, label='equ
    ax.set_title(f"{a=:.2f}, {eq_type}")
    ax.legend(loc="lower left")
```

```
Limit cycle found!
Limit cycle found!
```



Итог. Нейронная модель ФитцХью-Нагумо имеет одну точку бифуркации . При анализе системы была рассмотрена и визуализирована ее единственная равновесная точка. При изменении параметра , равновесная точка меняет тип 3 раза, а также после бифуркации в точке предельный цикл исчезает.

2. Климатическая модель Зальцман-Николис (SN)

1. Найдем равновесия системы.

2. Исследуем равновесия на устойчивость.

Далее с помощью численной процедуры исследуем поведение собственных чисел якобиана в равновесных точках при различных значениях параметра .

In [8]:

```
def model_SN(
    time: float,
    state: np.ndarray,
    a: float = 1,
    b: float = 2,
):
    """
    Saltzman-Nicolis climate model.
    Accepts `time` argument for interface consistency, but does not use it.

    Parameter restrictions:
    - a > 0
    - b = 2
    """
    x, y = state
    return np.array([
```

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```
        y - x,
        -a * x + b * y - (x ** 2) * y
    ])
```

In [9]:

```
SN = DynamicSystem2D(
    model_SN,
    params=dict(a=1, b=2)
)

state_init = np.array([0.1, 0.1])
time_span = np.linspace(0, 50, num=2000)
param_values = np.linspace(0, 3, num=50)
```

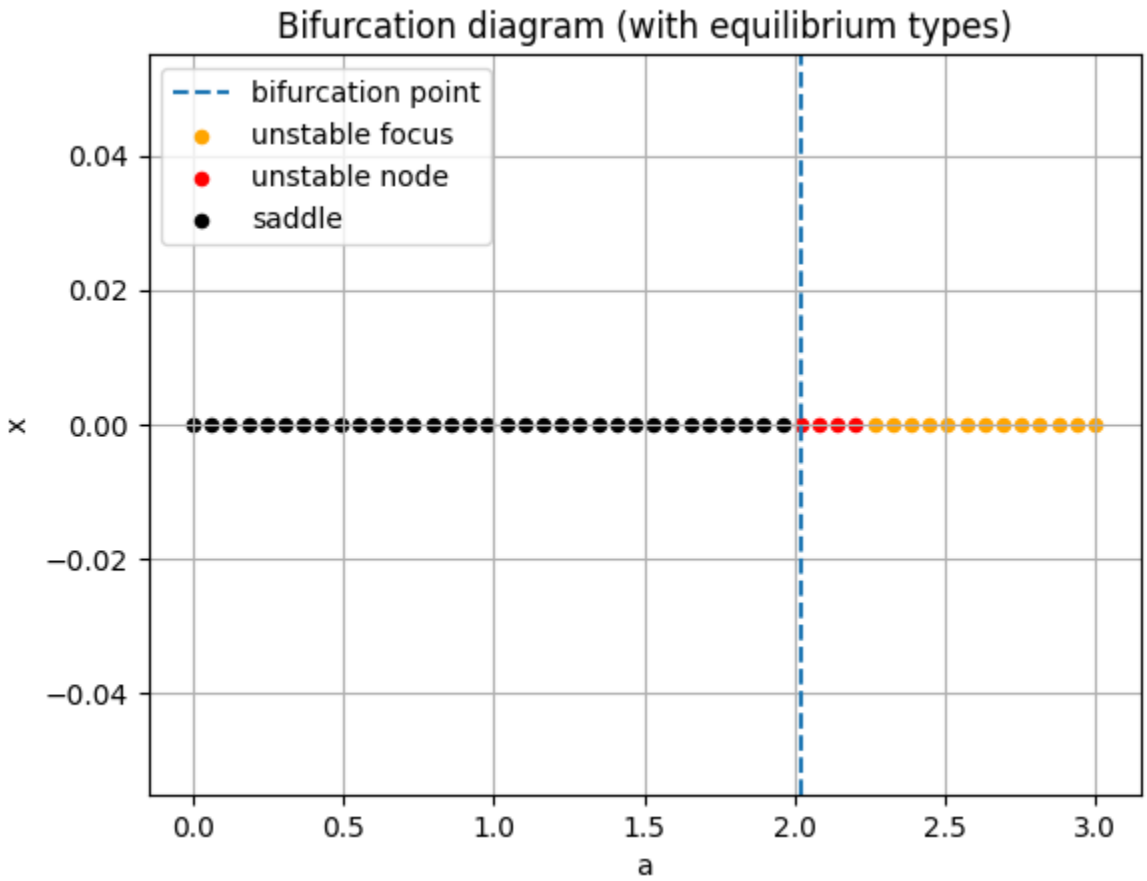
3. Построим бифуркационную диаграмму по параметру .

In [10]:

```
fig, ax = plt.subplots()

bifurcation_points, equilibria = SN.plot_bifurcation_diagram(
    param_name='a',
    param_values=param_values,
    state_init=state_init,
    ax=ax
)

a_bif = bifurcation_points[0]
```



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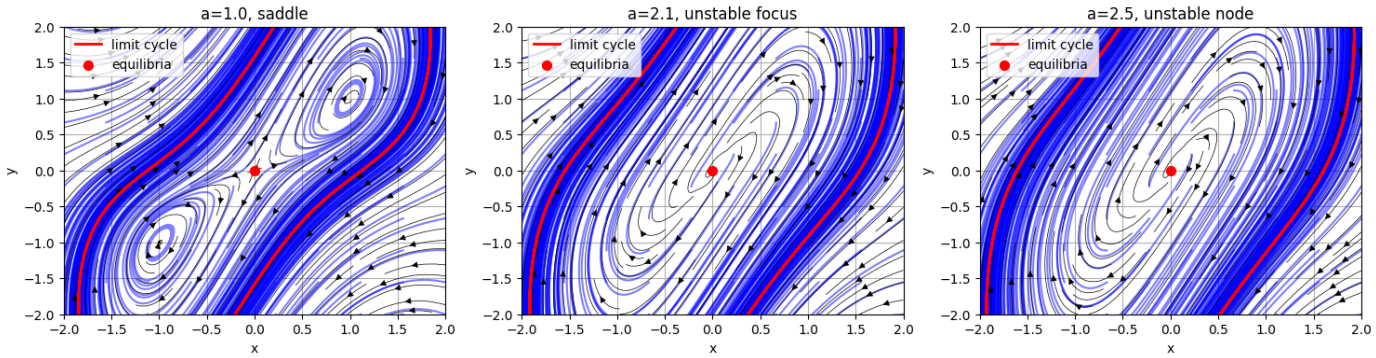
4. Построим типичные фазовые портреты для различных значений по бифуркационной диаграмме.

In [11]:

```
a_min = param_values[0]
a_max = param_values[-1]
a_values_to_types = [
    (1, "saddle"),
    (2.1, "unstable focus"),
    (2.5, "unstable node")
]
a_values, equilibrium_types = zip(*a_values_to_types)

equilibria_per_value = np.concatenate((param_values[:, None], equilibria), axis=1)
tolerance = abs(param_values[1] - param_values[0]) / 2

fig, axes = plt.subplots(1, len(a_values_to_types), figsize=(18, 4))

for ax, a, eq_type in zip(axes, a_values, equilibrium_types):
    SN.params['a'] = a
    SN.plot_phase_portrait_dense(
        ax=ax,
        time_span=time_span,
        bounds=(-2, 2),
        grid_size=10
    )

    equilibrium_idx = np.isclose(equilibria_per_value[:, 0], a, atol=tolerance)
    equilibrium_value = equilibria_per_value[equilibrium_idx][0]
    equilibrium_point = equilibrium_value[1:]

    ax.scatter(*equilibrium_point, color='red', marker='o', s=50, zorder=100, label='equ
    ax.set_title(f"{a=:.1f}, {eq_type}")
    ax.legend(loc="upper left")
```

```
Limit cycle found!
Limit cycle found!
Limit cycle found!
```



Итог. Климатическая модель Зальцман-Николис имеет одну точку бифуркации . При анализе системы была рассмотрена и визуализирована одна действительная равновесная точка из трех существующих. При изменении параметра , равновесная точка меняет тип 2 раза, а также меняется количество и форма предельных цклов.

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